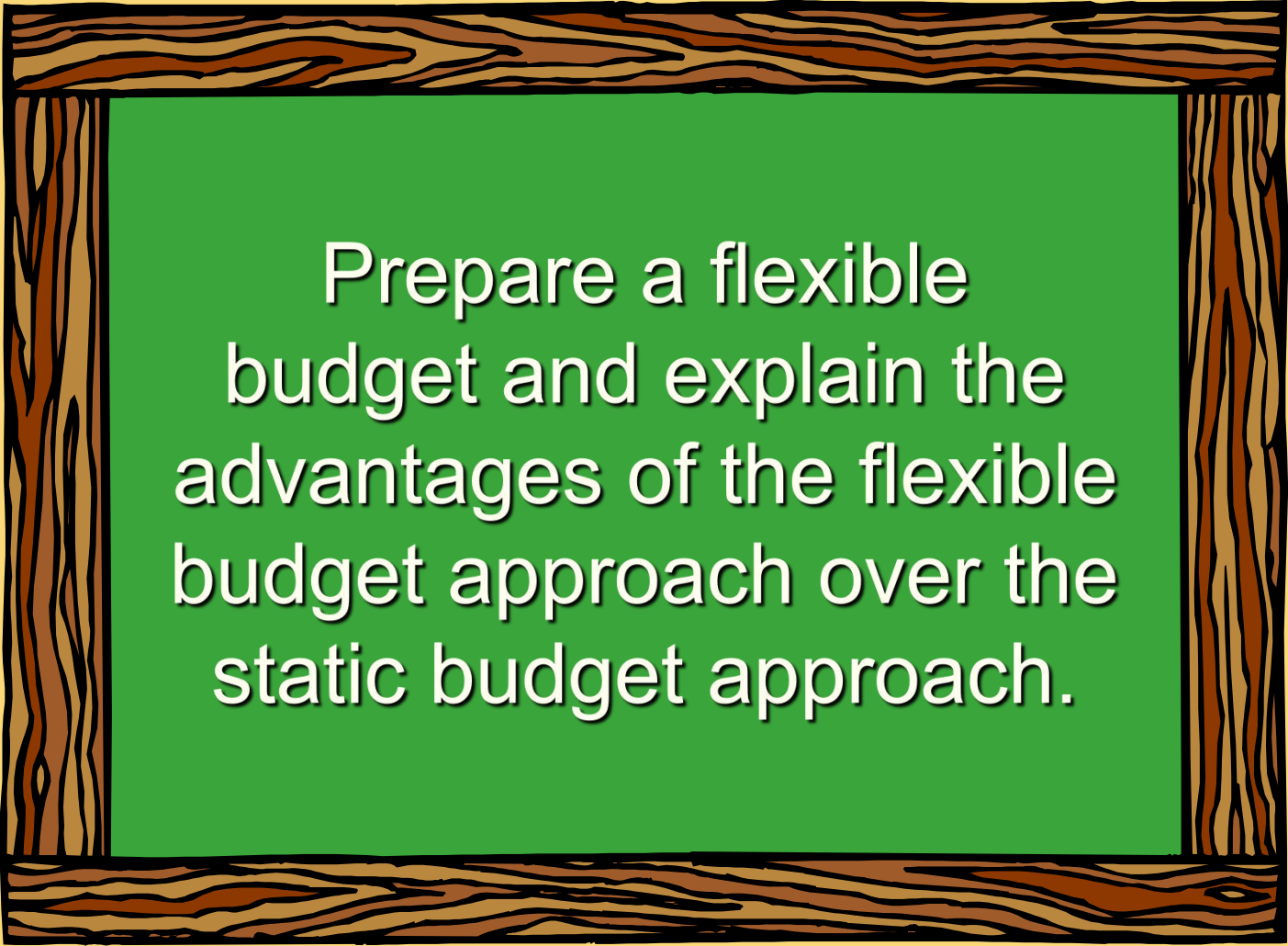


Flexible Budgets and Overhead Analysis

Chapter 11



Learning Objective 1

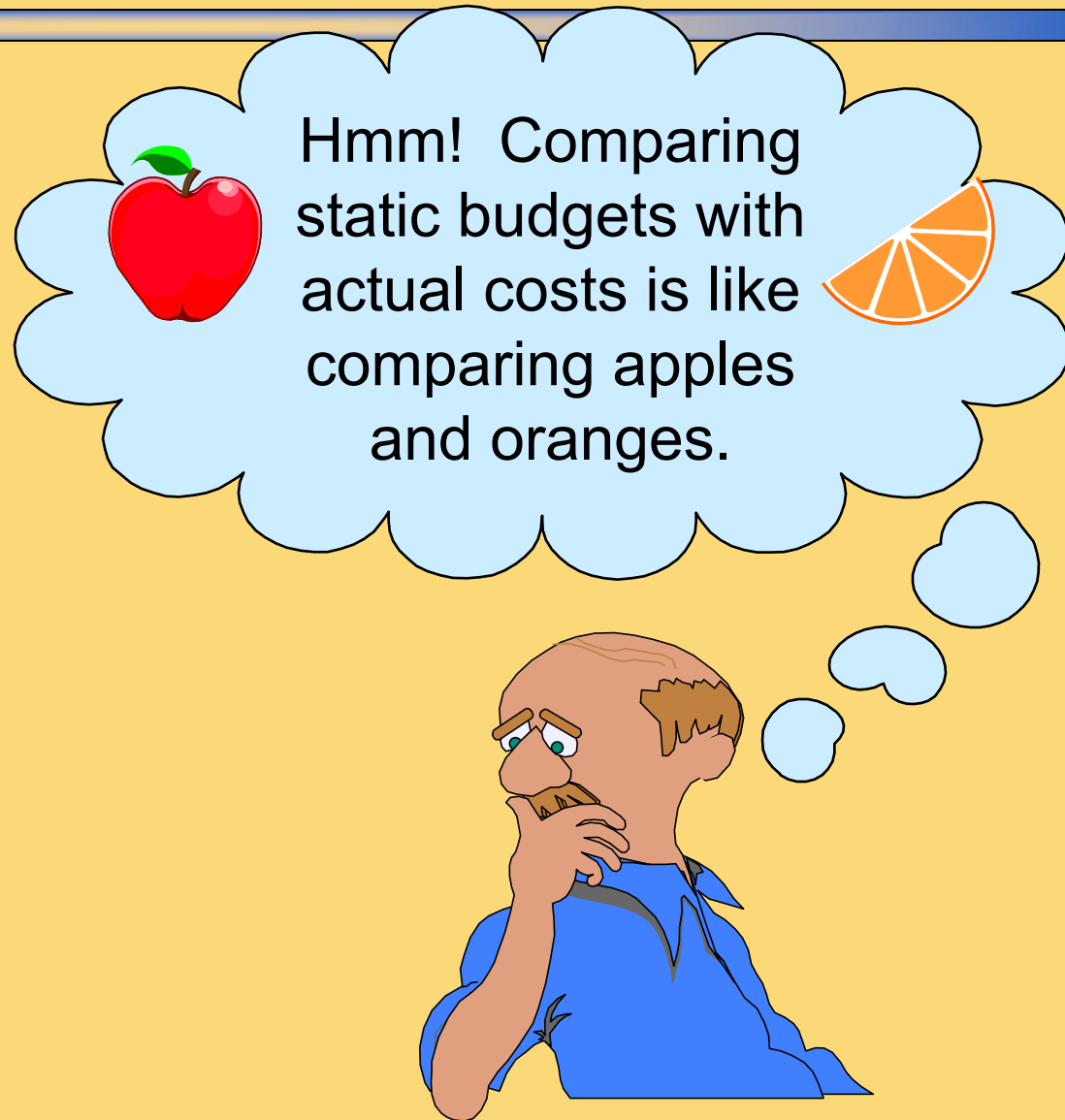


Prepare a flexible budget and explain the advantages of the flexible budget approach over the static budget approach.

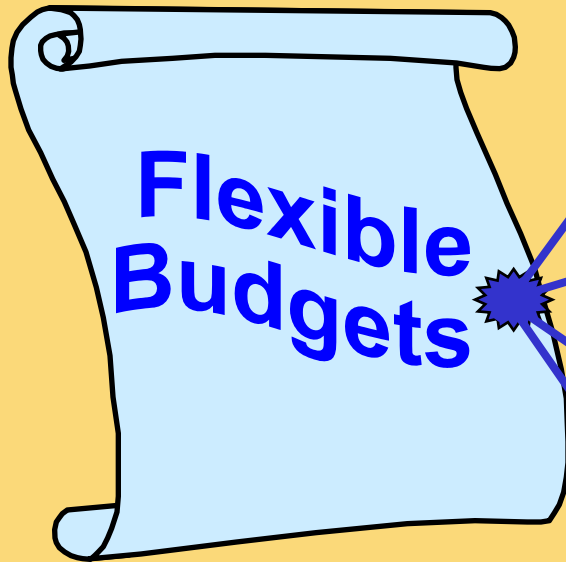
Static Budgets and Performance Reports

Static budgets are prepared for a single, **planned level** of activity.

Performance evaluation is difficult when actual activity differs from the planned level of activity.



Flexible Budgets



May be prepared for any activity level in the relevant range.

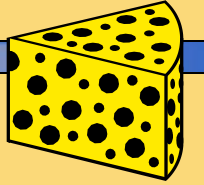
Show costs that should have been incurred at the actual level of activity, enabling “apples to apples” cost comparisons.

Reveal variances related to cost control.

Improve performance evaluation.

Let's look at CheeseCo.

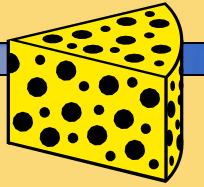
Static Budgets and Performance Reports



CheeseCo

	Static Budget	Actual Results	Variances
Machine hours	10,000		
Variable costs			
Indirect labor	\$ 40,000		
Indirect materials	30,000		
Power	5,000		
Fixed costs			
Depreciation	12,000		
Insurance	2,000		
Total overhead costs	\$ 89,000		

Static Budgets and Performance Reports



CheeseCo



**Static
Budget**



**Actual
Results**

Variances

Machine hours

10,000

8,000

Variable costs

Indirect labor

\$ 40,000

\$ 34,000

Indirect materials

30,000

25,500

Power

5,000

3,800

Fixed costs

Depreciation

12,000

12,000

Insurance

2,000

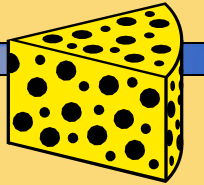
2,050

Total overhead costs

\$ 89,000

\$ 77,350

Static Budgets and Performance Reports



CheeseCo



Static
Budget



Actual
Results

Variances

Machine hours

10,000

8,000

2,000 U

Variable

Indirect

Indirect

Power

U = Unfavorable variance
**CheeseCo was unable to achieve
 the budgeted level of activity.**

\$6,000 F

4,500 F

1,200 F

Fixed costs

Depreciation

12,000

12,000

0

Insurance

2,000

2,050

50 U

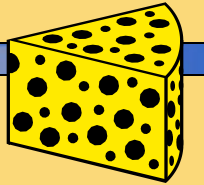
Total overhead costs

\$ 89,000

\$ 77,350

\$11,650 F

Static Budgets and Performance Reports

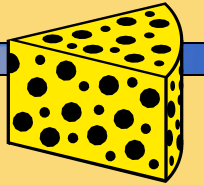


CheeseCo

	 Static Budget	 Actual Results	Variances
Machine hours	<u>10,000</u>	<u>8,000</u>	<u>2,000 U</u>
Variable costs			
Indirect labor	\$ 40,000	\$ 34,000	\$6,000 F
Indirect materials	30,000	25,500	4,500 F
Power	5,000	3,800	1,200 F
Insurance	<u>2,000</u>	<u>2,050</u>	<u>50 U</u>
Total overhead costs	<u>\$ 89,000</u>	<u>\$ 77,350</u>	<u>\$11,650 F</u>

F = Favorable variance that occurs when actual costs are less than budgeted costs.

Static Budgets and Performance Reports



CheeseCo

	 Static Budget	 Actual Results	Variances
Machine hours	<u>10,000</u>	<u>8,000</u>	<u>2,000 U</u>
Variable costs			
Indirect labor	\$ 40,000	\$ 34,000	\$6,000 F
Indirect materials	30,000	25,500	4,500 F
Power	5,000	3,800	1,200 F
Since cost variances are favorable, have we done a good job controlling costs?			
Insurance	<u>2,000</u>	<u>2,050</u>	<u>50 U</u>
Total overhead costs	<u>\$ 89,000</u>	<u>\$ 77,350</u>	<u>\$11,650 F</u>

Static Budgets and Performance Reports

I don't think I can answer the question using a static budget.

Actual activity is below budgeted activity.

So, shouldn't variable costs be lower if actual activity is lower?



Static Budgets and Performance Reports

- The relevant question is . . .

“How much of the favorable cost variance is due to lower activity, and how much is due to good cost control?”

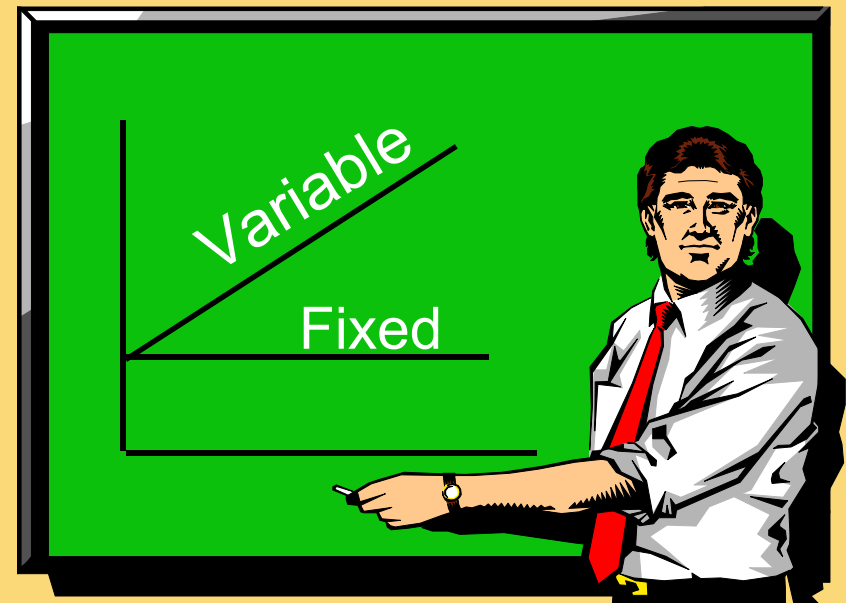
- To answer the question, we must **flex** the budget to the actual level of activity.



Preparing a Flexible Budget

To **flex** a budget we need to know that:

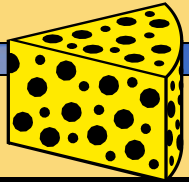
- ◆ **Total variable** costs **change** in direct proportion to changes in activity.
- ◆ **Total fixed** costs remain **unchanged** within the relevant range.



Preparing a Flexible Budget



Preparing a Flexible Budget



CheeseCo

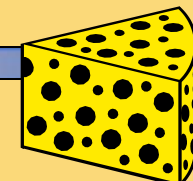
	Cost Formula per Hour	Total Fixed Cost	Flexible Budgets		
			8,000 Hours	10,000 Hours	12,000 Hours
Machine hours			8.000	10.000	12.000
Variable costs					
Indirect labor	\$ 4.00				
Indirect material	3.00				
Power	0.50				
Total variable cost	<u>\$ 7.50</u>				
Fixed costs					
Depreciation		\$ 12,000			
Insurance		2,000			
Total fixed cost					
Total overhead costs					

Variable costs are expressed as a constant amount per hour.

**$\$40,000 \div 10,000$ hours is
\$4.00 per hour.**

Fixed costs are expressed as a total amount.

Preparing a Flexible Budget

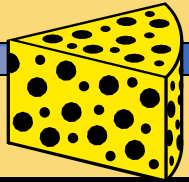


CheeseCo

	Cost Formula per Hour	Total Fixed Cost	Flexible Budgets		
			8,000 Hours	10,000 Hours	12,000 Hours
Machine hours			8,000	10,000	12,000
Variable costs					
Indirect labor	\$ 4.00		\$ 32,000		
Indirect material	3.00		24,000		
Power	0.50		4,000		
Total variable cost	\$ 7.50		\$ 60,000		
Fixed costs					
Depreciation					
Insurance		2,000			
Total fixed cost					
Total overhead costs					

$$\$4.00 \text{ per hour} \times 8,000 \text{ hours} = \$32,000$$

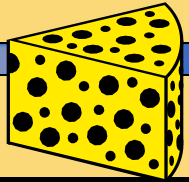
Preparing a Flexible Budget



CheeseCo

	Cost Formula per Hour	Total Fixed Cost	Flexible Budgets		
			8,000 Hours	10,000 Hours	12,000 Hours
Machine hours			8,000	10,000	12,000
Variable costs					
Indirect labor	\$ 4.00		\$ 32,000		
Indirect material	3.00		24,000		
Power	0.50		4,000		
Total variable cost	<u>\$ 7.50</u>		<u>\$ 60,000</u>		
Fixed costs					
Depreciation		\$ 12,000	\$ 12,000		
Insurance		2,000	2,000		
Total fixed cost			<u>\$ 14,000</u>		
Total overhead costs			<u>\$ 74,000</u>		

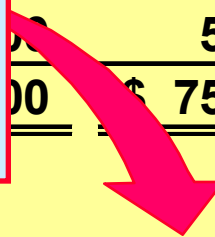
Preparing a Flexible Budget



CheeseCo

	Cost Formula per Hour	Total Fixed Cost	Flexible Budgets		
			8,000 Hours	10,000 Hours	12,000 Hours
Machine hours			8,000	10,000	12,000
Variable costs					
Indirect labor	\$ 4.00		\$ 32,000	\$ 40,000	
Indirect material			00	30,000	
Power			00	5,000	
Total variable cost	\$		00	\$ 75,000	
Fixed costs					
Depreciation		\$ 12,000	\$ 12,000	\$ 12,000	
Insurance		2,000	2,000	2,000	
Total fixed cost			\$ 14,000	\$ 14,000	
Total overhead costs			\$ 74,000	\$ 89,000	?

**Total fixed costs
do not change in
the relevant range.**



Quick Check ✓

What should be the total overhead costs for the Flexible Budget at 12,000 hours?

- a. \$92,500.
- b. \$89,000.
- c. \$106,800.
- d. \$104,000.

Quick Check ✓

What should be the total overhead costs for the Flexible Budget at 12,000 hours?

- a. \$92,500.
- b. \$89,000.
- c. \$106,800.
- d. \$104,000.

Total overhead cost

$$= \$14,000 + \$7.50 \text{ per hour} \times 12,000 \text{ hours}$$

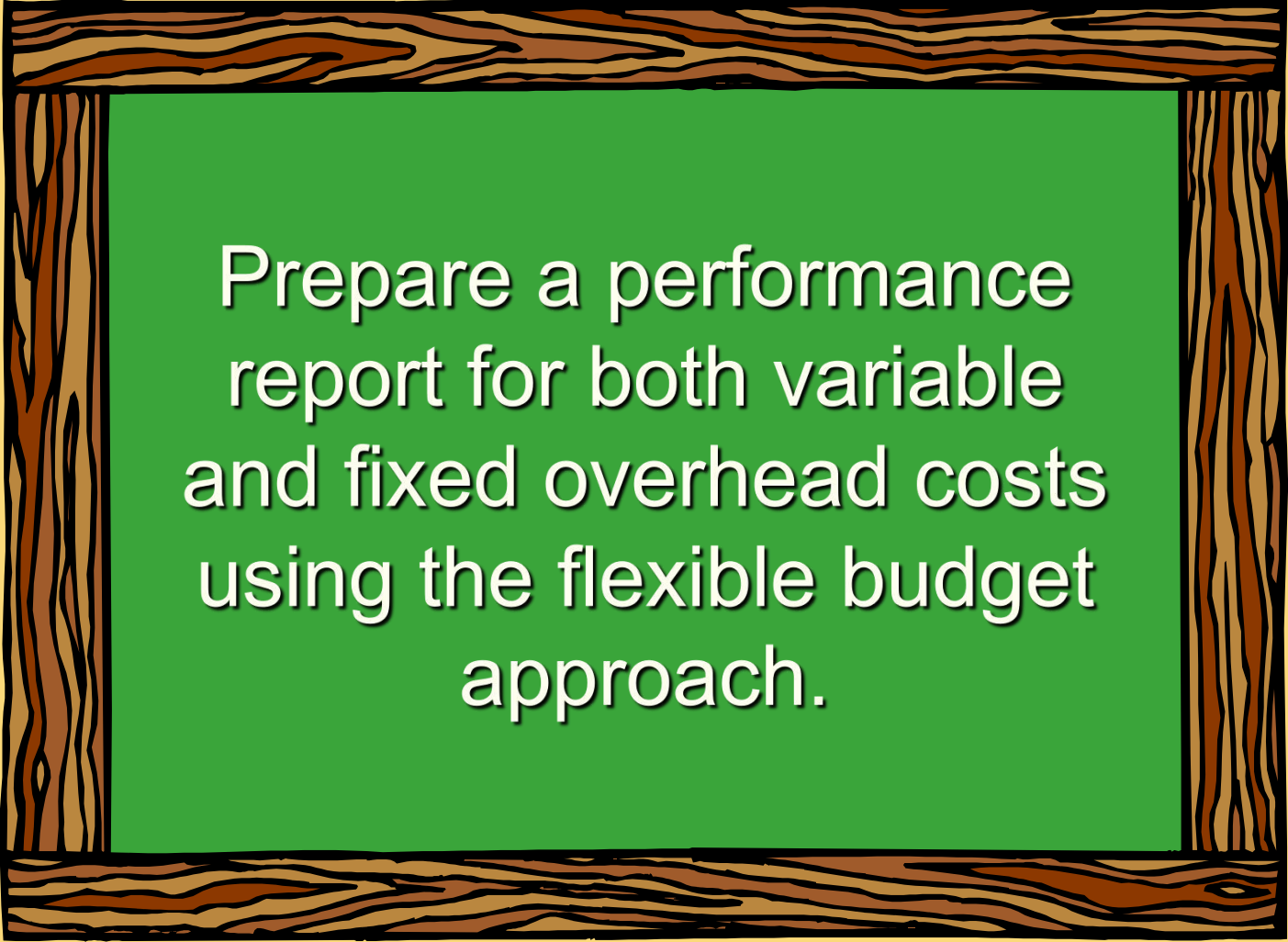
$$= \$14,000 + \$90,000 = \$104,000$$

Preparing a Flexible Budget



	Cost Formula per Hour	Total Fixed Cost	Flexible Budgets		
			8,000 Hours	10,000 Hours	12,000 Hours
Machine hours			8,000	10,000	12,000
Variable costs					
Indirect labor	\$ 4.00		\$ 32,000	\$ 40,000	\$ 48,000
Indirect material	3.00		24,000	30,000	36,000
Power	0.50		4,000	5,000	6,000
Total variable cost	<u>\$ 7.50</u>		<u>\$ 60,000</u>	<u>\$ 75,000</u>	<u>\$ 90,000</u>
Fixed costs					
Depreciation		\$ 12,000	\$ 12,000	\$ 12,000	\$ 12,000
Insurance		2,000	2,000	2,000	2,000
Total fixed cost			<u>\$ 14,000</u>	<u>\$ 14,000</u>	<u>\$ 14,000</u>
Total overhead costs			<u>\$ 74,000</u>	<u>\$ 89,000</u>	<u>\$ 104,000</u>

Learning Objective 2



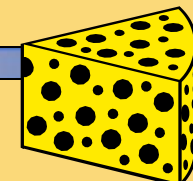
Prepare a performance report for both variable and fixed overhead costs using the flexible budget approach.

Flexible Budget Performance Report

Let's prepare a **flexible**
budget performance report
for CheeseCo.



Flexible Budget Performance Report

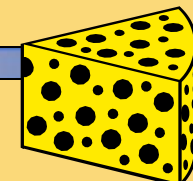


CheeseCo

Flexible budget is prepared for the same activity level (8,000 hours) as actually achieved.

		la ur	Total Fixed Cost	 Flexible Budget	 Actual Results	Variances
M				8,000	8,000	0
V						
	Indirect labor	\$ 4.00			\$ 34,000	
	Indirect material	3.00			25,500	
	Power	0.50			3,800	
	Total variable cost	<u>\$ 7.50</u>			<u>\$ 63,300</u>	
	Fixed costs					
	Depreciation		\$ 12,000		\$ 12,000	
	Insurance		2,000		2,050	
	Total fixed cost				<u>\$ 14,050</u>	
	Total overhead costs				<u>\$ 77,350</u>	

Flexible Budget Performance Report



CheeseCo

	Cost Formula per Hour	Total Fixed Cost	 Flexible Budget	 Actual Results	Variances
Machine hours			8,000	8,000	0
Variable costs					
Indirect labor	\$ 4.00		\$ 32,000	\$ 34,000	\$ 2,000 U
Indirect material	3.00		24,000	25,500	1,500 U
Power	0.50		4,000	3,800	200 F
Total variable cost	<u>\$ 7.50</u>		<u>\$ 60,000</u>	<u>\$ 63,300</u>	<u>\$ 3,300 U</u>
Fixed costs					
Depreciation		\$ 12,000	\$ 12,000	\$ 12,000	\$ 0
Insurance		2,000	2,000	2,050	50 U
Total fixed cost			<u>\$ 14,000</u>	<u>\$ 14,050</u>	<u>50 U</u>
Total overhead costs			<u>\$ 74,000</u>	<u>\$ 77,350</u>	<u>\$ 3,350 U</u>

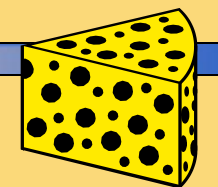
Flexible Budget Performance Report

Remember the question:
“How much of the total
variance is due to lower
activity and how much is
due to cost control?”



Static Budgets and Performance

How much of the \$11,650 favorable variance is due to lower activity and how much is due to cost control?



**Static
Budget**



**Actual
Results**

Variances

Machine hours

10,000

8,000

2,000 U

Variable costs

Indirect labor

\$ 40,000

\$ 34,000

\$6,000 F

Indirect materials

30,000

25,500

4,500 F

Power

5,000

3,800

1,200 F

Fixed costs

Depreciation

12,000

12,000

0

Insurance

2,000

2,050

50 U

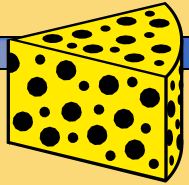
Total overhead costs

\$ 89,000

\$ 77,350

\$11,650 F

Flexible Budget Performance Report



Overhead Variance Analysis



**Static
Overhead
Budget at
10,000 Hours**

\$ 89,000

**Let's place
the flexible
budget for
8,000 hours
here.**

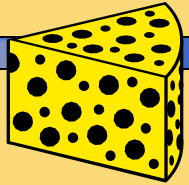


**Actual
Overhead
at
8,000 Hours**

\$ 77,350

**Difference between original static budget
and actual overhead = \$11,650 F.**

Flexible Budget Performance Report



Overhead Variance Analysis

 Static Overhead Budget at <u>10,000 Hours</u>	 Flexible Overhead Budget at <u>8,000 Hours</u>	 Actual Overhead at <u>8,000 Hours</u>
\$ 89,000	\$ 74,000	\$ 77,350

Activity

Cost control

This \$15,000**F** variance is
due to lower activity.

This \$3,350**U**
variance is due
to poor cost control.

Variable Overhead Variances – A Closer Look

If flexible budget
is based on
actual hours

Only a **spending**
variance can be
computed.

If flexible budget
is based on both
actual hours and
standard hours

Both **spending**
and **efficiency**
variances can be
computed.

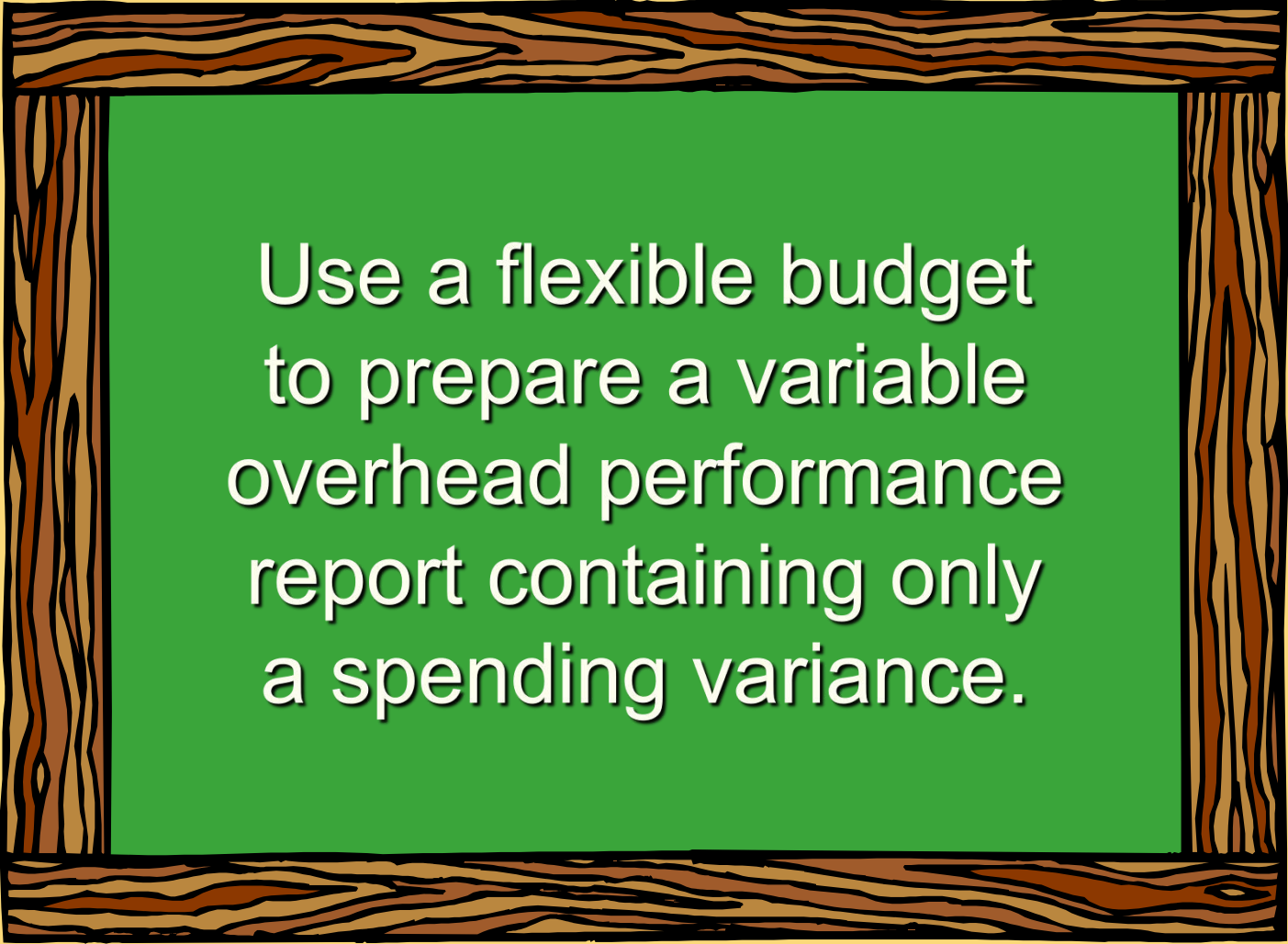
Variable Overhead Variances – Example

- ColaCo's actual production for the period required 3,200 standard machine hours.
- Actual variable overhead incurred for the period was \$6,740.
- Actual machine hours worked were 3,300.
- The standard variable overhead cost per machine hour is \$2.00.

Compute the variable overhead spending variance first using actual hours. Then use standard hours allowed to calculate the variable overhead efficiency variance.



Learning Objective 3



Use a flexible budget
to prepare a variable
overhead performance
report containing only
a spending variance.

Variable Overhead Variances



**Actual
Variable
Overhead
Incurred**

$AH \times AR$

**Flexible Budget
for Variable
Overhead at
Actual Hours**

$AH \times SR$

**Spending
Variance**

AH = Actual hours
**AR = Actual variable
overhead rate**
**SR = Standard variable
overhead rate**

Spending variance = $AH(AR - SR)$

Variable Overhead Variances – Example



**Actual
Variable
Overhead
Incurred**

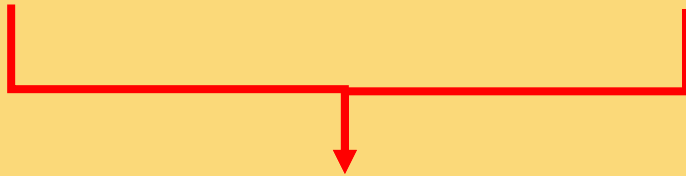
**Flexible Budget
for Variable
Overhead at
Actual Hours**

3,300 hours

×

**\$2.00 per hour
= \$6,600**

\$6,740



**Spending Variance
= \$140 unfavorable**

Variable Overhead Variances – A Closer Look

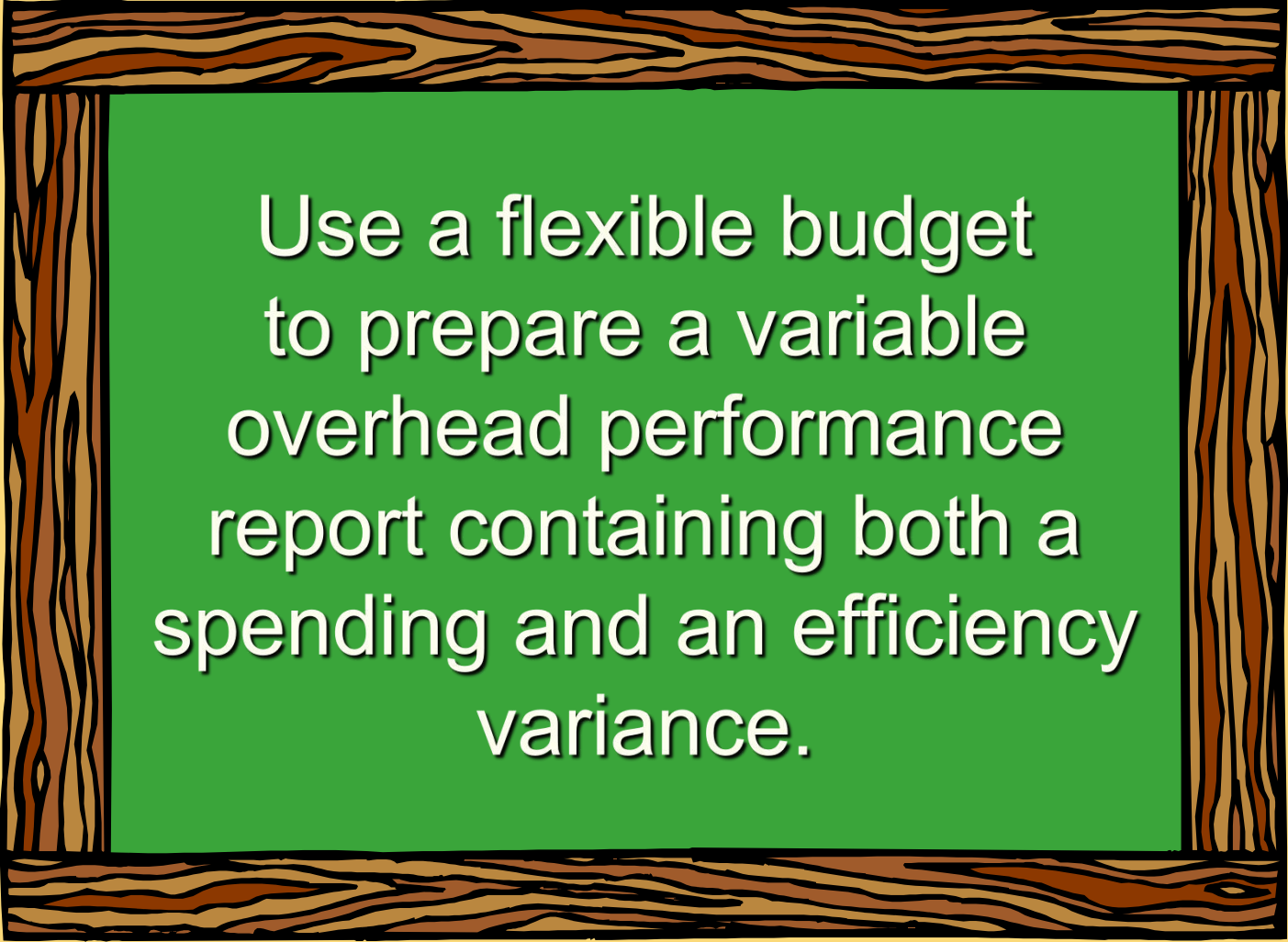
Spending Variance

Results from paying more or less than expected for overhead items and from excessive usage of overhead items.



Now, let's use the standard hours allowed, along with the actual hours, to compute the efficiency variance.

Learning Objective 4



Use a flexible budget
to prepare a variable
overhead performance
report containing both a
spending and an efficiency
variance.

Variable Overhead Variances

**Actual
Variable
Overhead
Incurred**

$AH \times AR$

**Flexible Budget
for Variable
Overhead at
Actual Hours**

**$AH \times$
 SR**

**Flexible Budget
for Variable
Overhead at
Standard Hours**

**$SH \times$
 SR**

**Spending
Variance**

**Efficiency
Variance**

Spending variance = $AH(AR - SR)$

Efficiency variance = $SR(AH - SH)$

Variable Overhead Variances – Example



**Actual
Variable
Overhead
Incurred**

**Flexible Budget
for Variable
Overhead at
Actual Hours**

**Flexible Budget
for Variable
Overhead at
Standard Hours**

3,300 hours

×

\$2.00 per hour

3,200 hours

×

\$2.00 per hour

\$6,740

\$6,600

\$6,400

Spending variance
\$140 unfavorable

Efficiency variance
\$200 unfavorable

\$340 unfavorable flexible budget total variance

Variable Overhead Variances – A Closer Look

Efficiency Variance

Controlled by
managing the
overhead cost driver.



Quick Check ✓

Yoder Enterprises' actual production for the period required 2,100 standard direct labor hours. Actual variable overhead for the period was \$10,950. Actual direct labor hours worked were 2,050. The predetermined variable overhead rate is \$5 per direct labor hour. What was the spending variance?

- a. \$450 U
- b. \$450 F
- c. \$700 F
- d. \$700 U

Quick Check ✓

Spending variance = AH (AR - SR)

= Actual variable overhead incurred – (AH × SR)

= \$10,950 – (2,050 hours × \$5 per hour)

= \$10,950 – \$10,250

= \$700 U

was the spending variance:

a. \$450 U

b. \$450 F

c. \$700 F

d. \$700 U

Quick Check ✓

Yoder Enterprises' actual production for the period required 2,100 standard direct labor hours. Actual variable overhead for the period was \$10,950. Actual direct labor hours worked were 2,050. The predetermined variable overhead rate is \$5 per direct labor hour. What was the efficiency variance?

- a. \$450 U
- b. \$450 F
- c. \$250 F
- d. \$250 U

Quick Check ✓

Yoder Enterprises' actual production for the period required 2,100 standard direct labor hours. Actual variable overhead for the period

$$\text{Efficiency variance} = \text{SR} (\text{AH} - \text{SH})$$

$$= \$5 \text{ per hour} (2,050 \text{ hours} - 2,100 \text{ hours})$$

$$= \$250 \text{ F}$$

a. \$450 U

b. \$450 F

c. \$250 F

d. \$250 U

Quick Check Summary

**Actual
Variable
Overhead
Incurred**

\$10,950

**Flexible Budget
for Variable
Overhead at
Actual Hours**

2,050 hours

×

\$5 per hour

\$10,250

**Flexible Budget
for Variable
Overhead at
Standard Hours**

2,100 hours

×

\$5 per hour

\$10,500

Spending variance
\$700 unfavorable

Efficiency variance
\$250 favorable

\$450 unfavorable flexible budget total variance

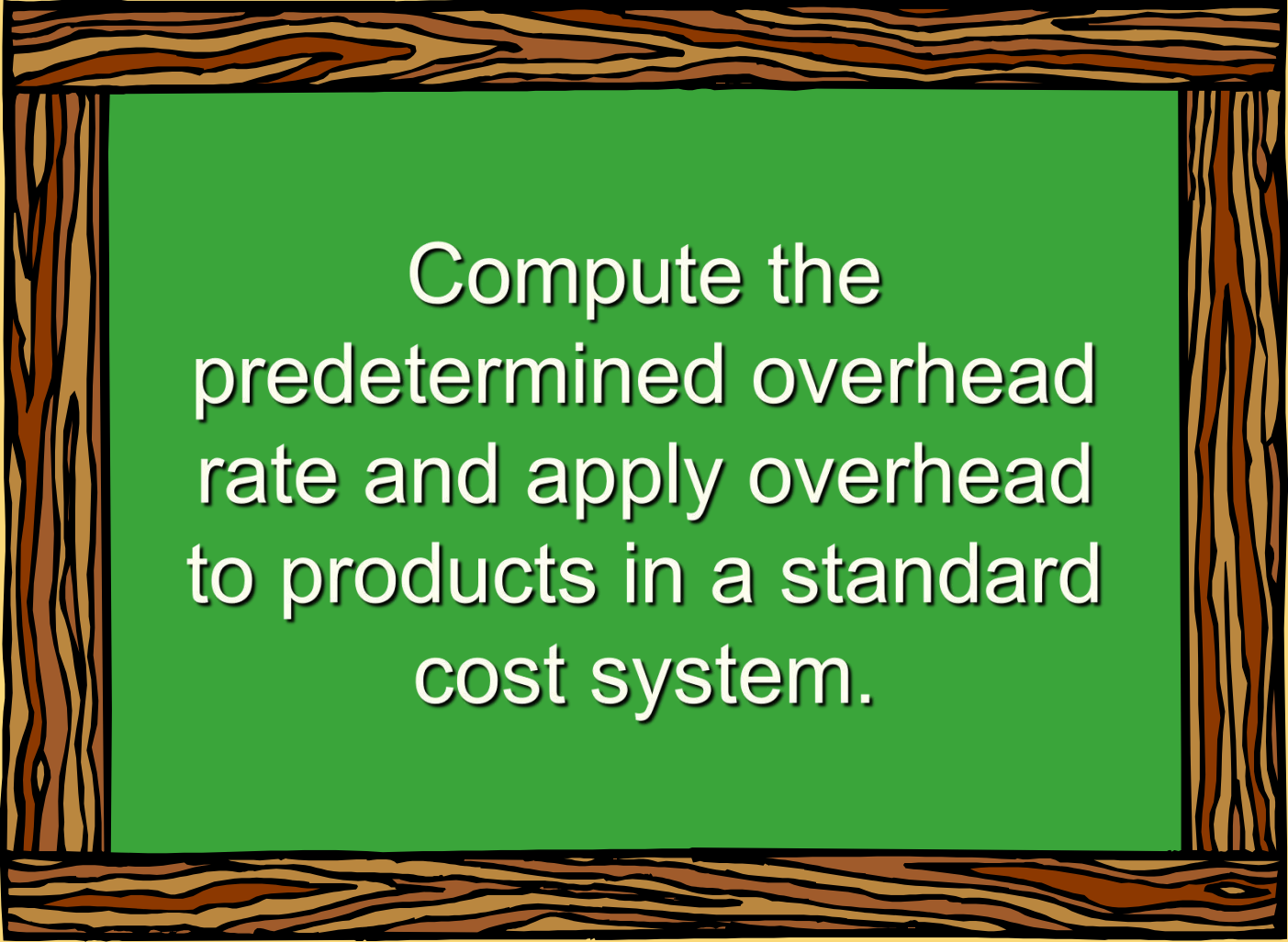
Activity-based Costing and the Flexible Budget

It is unlikely that all variable overhead will be driven by a single activity.



Activity-based costing can be used when multiple activity bases drive variable overhead costs.

Learning Objective 5

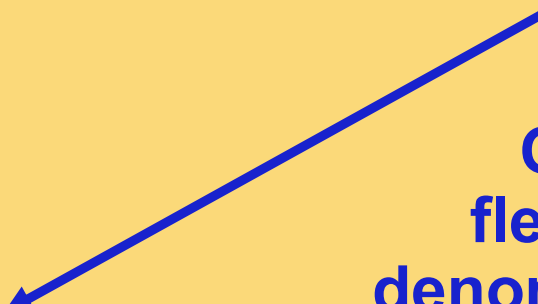


Compute the
predetermined overhead
rate and apply overhead
to products in a standard
cost system.

Overhead Rates and Overhead Analysis

Recall that overhead costs are assigned to products and services using a **predetermined overhead rate (POHR)**:

Assigned Overhead = POHR × Standard Activity


$$\text{POHR} = \frac{\text{Overhead from the flexible budget for the denominator level of activity}}{\text{Denominator level of activity}}$$

Overhead Rates and Overhead Analysis

**The predetermined overhead rate
can be broken down into fixed
and variable components.**

The variable
component is useful
for preparing and analyzing
variable overhead
variances.

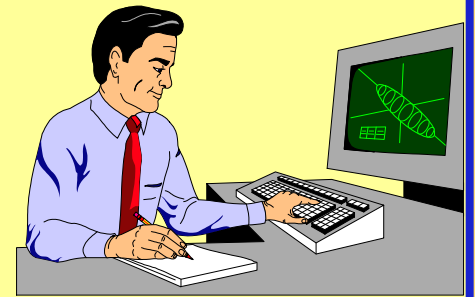
The fixed
component is useful
for preparing and analyzing
fixed overhead
variances.

Normal versus Standard Cost Systems

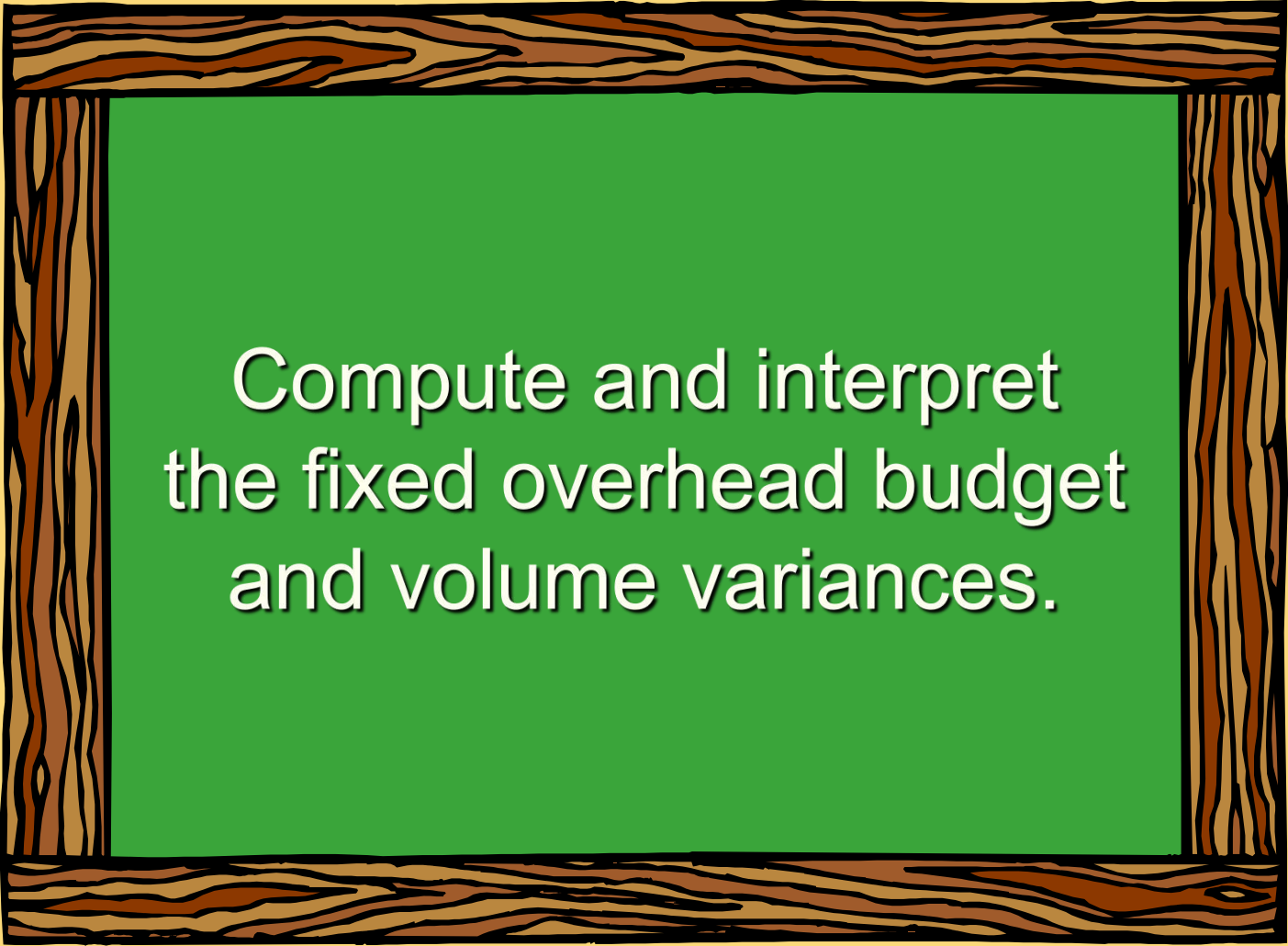
In a **normal cost** system, overhead is applied to work in process based on the actual number of hours worked in the period.



In a **standard cost** system, overhead is applied to work in process based on the standard hours allowed for the actual output of the period.

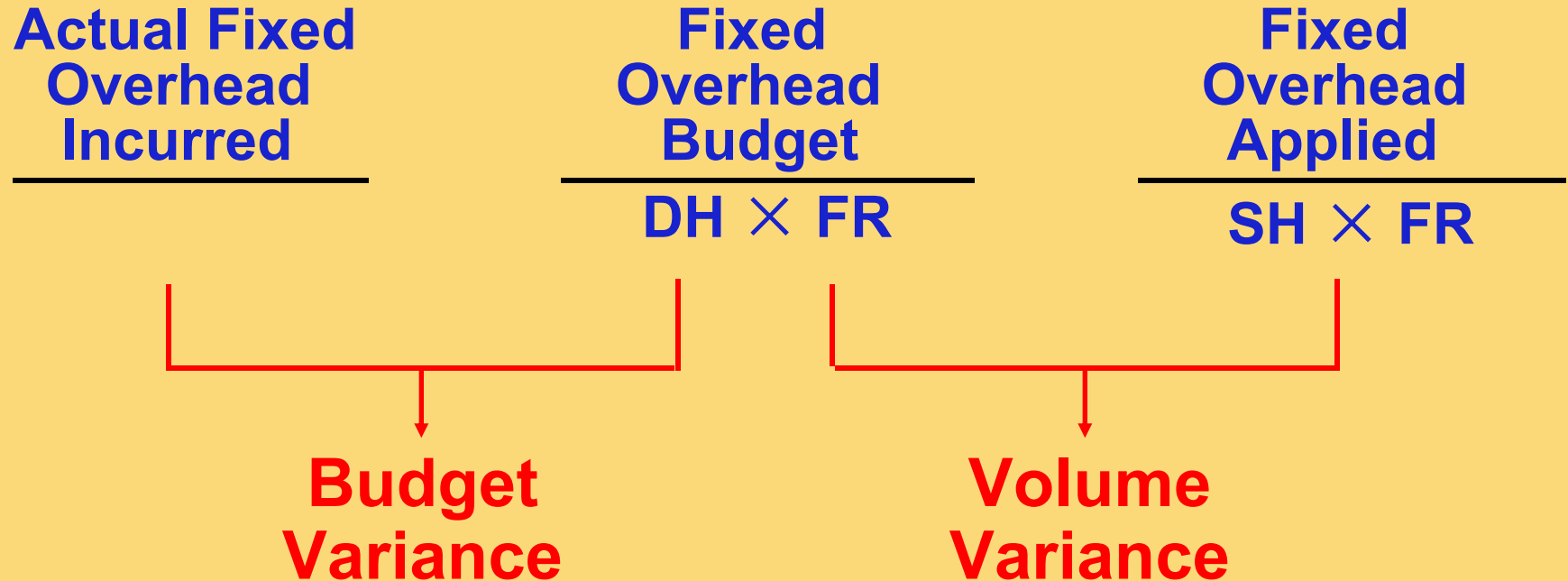


Learning Objective 6



Compute and interpret
the fixed overhead budget
and volume variances.

Fixed Overhead Variances



FR = Standard Fixed Overhead Rate
SH = Standard Hours Allowed
DH = Denominator Hours

Overhead Rates and Overhead Analysis – Example



ColaCo prepared this **flexible** budget for overhead:

Machine Hours	Total Variable Overhead	Variable Overhead Rate	Total Fixed Overhead	Fixed Overhead Rate
3,000	\$ 6,000	?	\$ 9,000	?
4,000	8,000	?	9,000	?

Let's calculate overhead rates.

ColaCo applies overhead based on machine-hour activity.

Overhead Rates and Overhead Analysis – Example



ColaCo prepared this **flexible** budget for overhead:

Machine Hours	Total Variable Overhead	Variable Overhead Rate	Total Fixed Overhead	Fixed Overhead Rate
3,000	\$ 6,000	\$ 2.00	\$ 9,000	?
4,000	8,000	2.00	9,000	?

Rate = Total Variable Overhead ÷ Machine Hours

This rate is constant at all levels of activity.

Overhead Rates and Overhead Analysis – Example



ColaCo prepared this **flexible** budget for overhead:

Machine Hours	Total Variable Overhead	Variable Overhead Rate	Total Fixed Overhead	Fixed Overhead Rate
3,000	\$ 6,000	\$ 2.00	\$ 9,000	\$ 3.00
4,000	8,000	2.00	9,000	2.25

Rate = Total Fixed Overhead ÷ Machine Hours

This rate decreases when activity increases.

Overhead Rates and Overhead Analysis – Example



ColaCo prepared this **flexible** budget for overhead:

Machine Hours	Total Variable Overhead	Variable Overhead Rate	Total Fixed Overhead	Fixed Overhead Rate
3,000	\$ 6,000	\$ 2.00	\$ 9,000	\$ 3.00
4,000	8,000	2.00	9,000	2.25

The total POHR is the sum of the fixed and variable rates for a given activity level.

Fixed Overhead Variances – Example

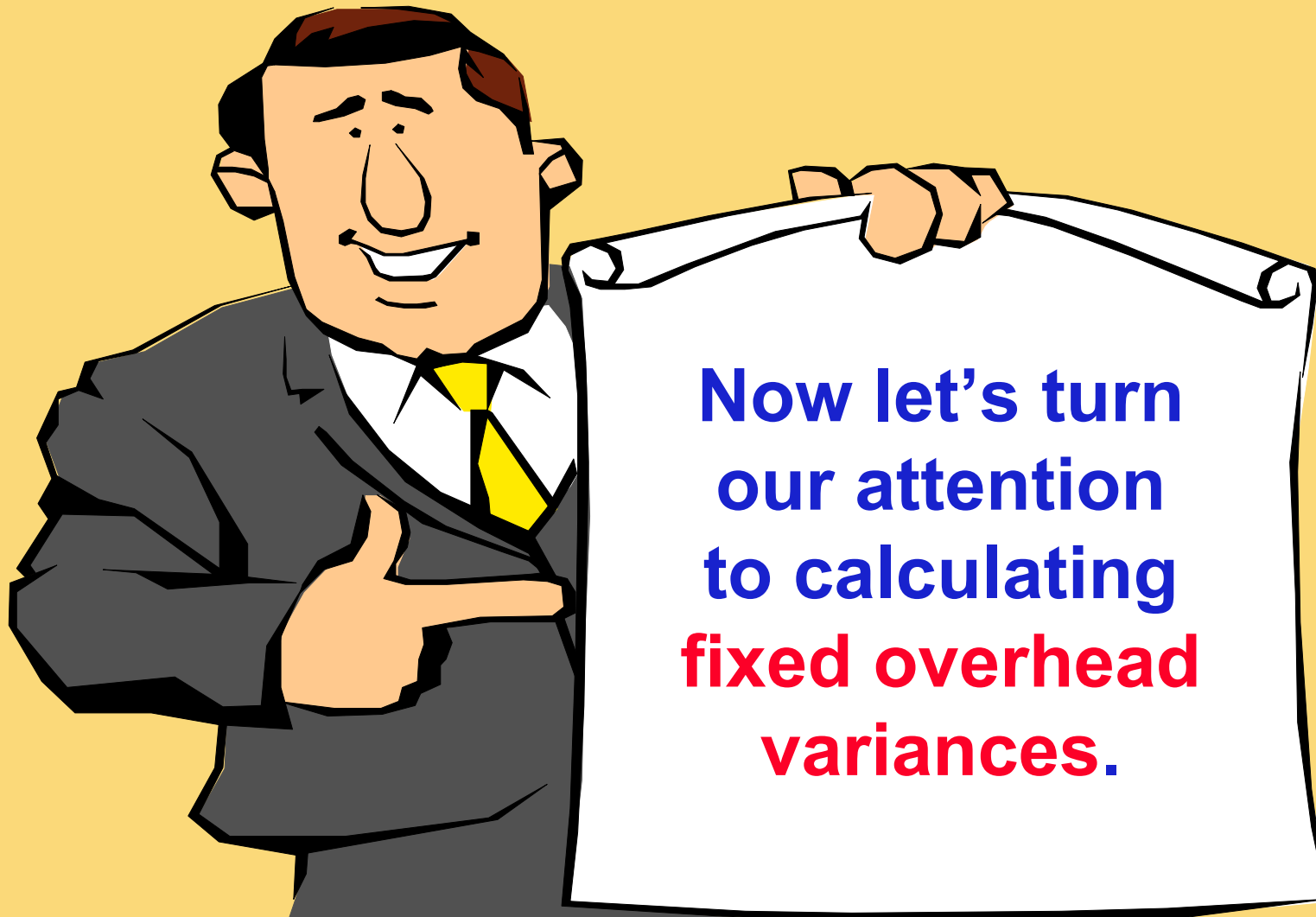


ColaCo's actual production required 3,200 **standard** machine hours. Actual fixed overhead was \$8,450. The predetermined overhead rate is based on 3,000 machine hours.



Which activity level you will chose 3,000 or 4000 MH?

Overhead Variances



Fixed Overhead Variances – Example



**Actual Fixed
Overhead
Incurred**

**Fixed
Overhead
Budget**

**Fixed
Overhead
Applied**

\$8,450

\$9,000

Budget variance
\$550 favorable

Fixed Overhead Variances – A Closer Look

Budget Variance

Results from spending more or less than expected for fixed overhead items.



Now, let's use the standard hours allowed to compute the fixed overhead volume variance.

Fixed Overhead Variances – Example



**Actual Fixed
Overhead
Incurred**

**Fixed
Overhead
Budget**

**Fixed
Overhead
Applied**

SH × FR

3,200 hours

×

\$3.00 per hour

\$8,450

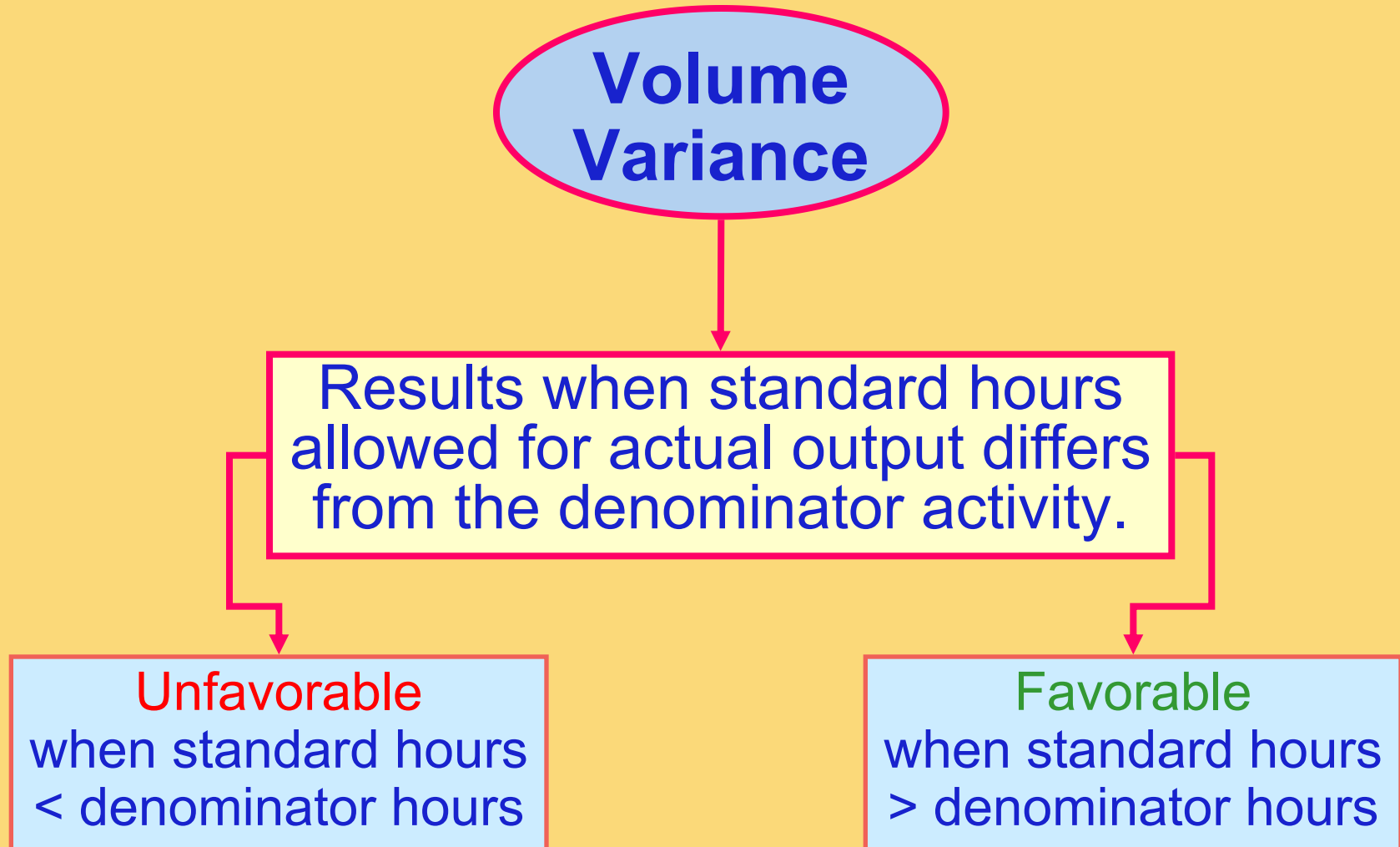
\$9,000

\$9,600

Budget variance
\$550 favorable

Volume variance
\$600 favorable

Volume Variance – A Closer Look



Volume Variance – A Closer Look

**Does not measure over-
or under spending**

**It results from treating fixed
overhead as if it were a
variable cost.**

Unfavorable
when standard hours
< denominator hours

Favorable
when standard hours
> denominator hours

Quick Check ✓

Yoder Enterprises' actual production for the period required 2,100 standard direct labor hours. Actual fixed overhead for the period was \$14,800. The budgeted fixed overhead was \$14,450. The predetermined fixed overhead rate was \$7 per direct labor hour. What was the budget variance?

- a. \$350 U
- b. \$350 F
- c. \$100 F
- d. \$100 U

Quick Check ✓

Budget variance

= Actual fixed overhead – Budgeted fixed overhead

= \$14,800 – \$14,450

= \$350 U

overhead rate was \$7 per direct labor hour.
What was the budget variance?

a. \$350 U

b. \$350 F

c. \$100 F

d. \$100 U

Quick Check ✓

Yoder Enterprises' actual production for the period required 2,100 standard direct labor hours. Actual fixed overhead for the period was \$14,800. The budgeted fixed overhead was \$14,450. The predetermined fixed overhead rate was \$7 per direct labor hour. What was the volume variance?

- a. \$250 U
- b. \$250 F
- c. \$100 F
- d. \$100 U

Quick Check ✓

Volume variance

$$= \text{Budgeted fixed overhead} - (\text{SH} \times \text{FR})$$

$$= \$14,450 - (2,100 \text{ hours} \times \$7 \text{ per hour})$$

$$= \$14,450 - \$14,700$$

$$= \$250 \text{ F}$$

overhead rate was \$7 per direct labor hour.
What was the volume variance?

a. \$250 U

b. \$250 F

c. \$100 F

d. \$100 U

Quick Check Summary

**Actual Fixed
Overhead
Incurred**

**Fixed
Overhead
Budget**

**Fixed
Overhead
Applied**

SH × FR

2,100 hours

×

\$7.00 per hour

\$14,800

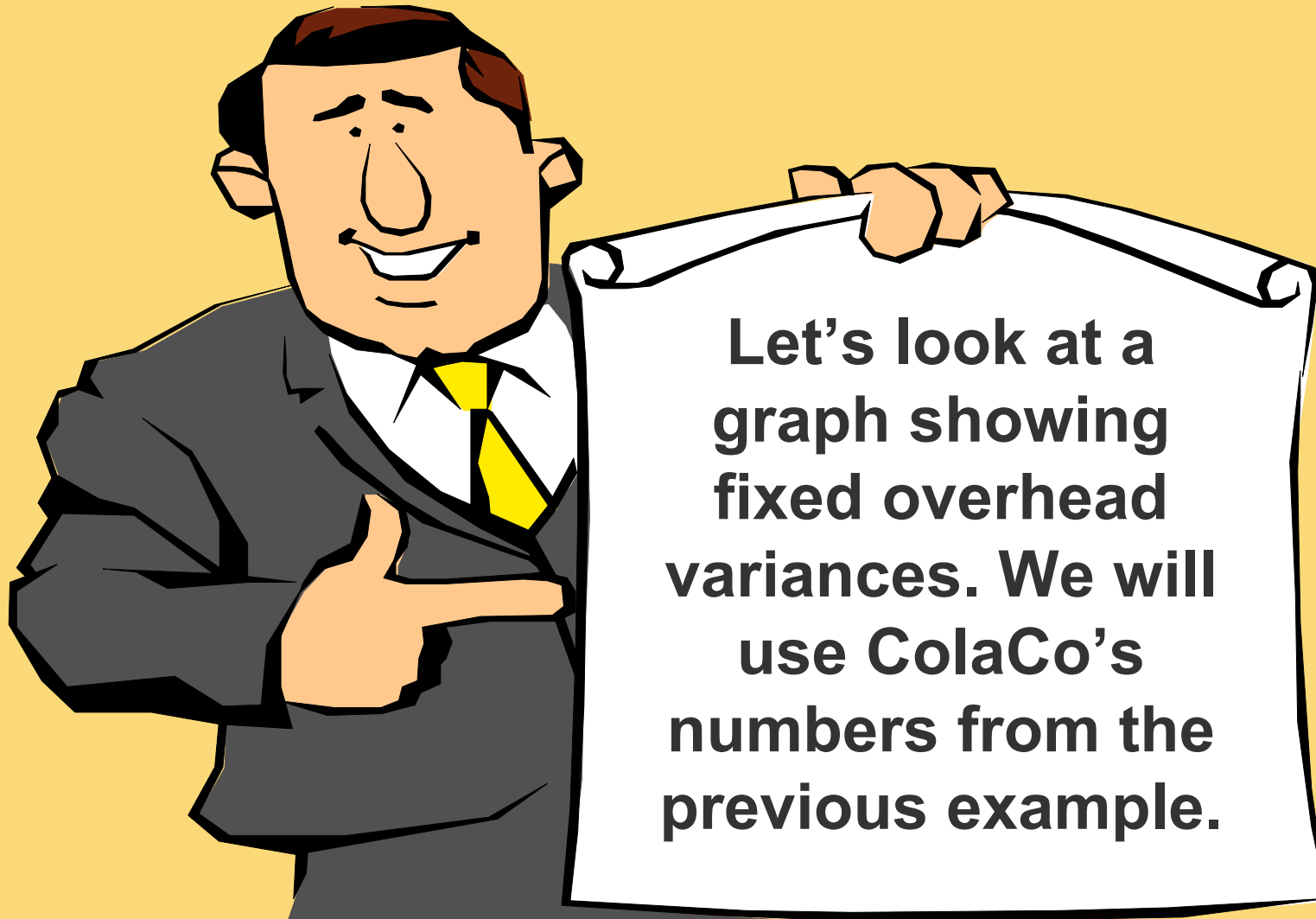
\$14,450

\$14,700

Budget variance
\$350 unfavorable

Volume variance
\$250 favorable

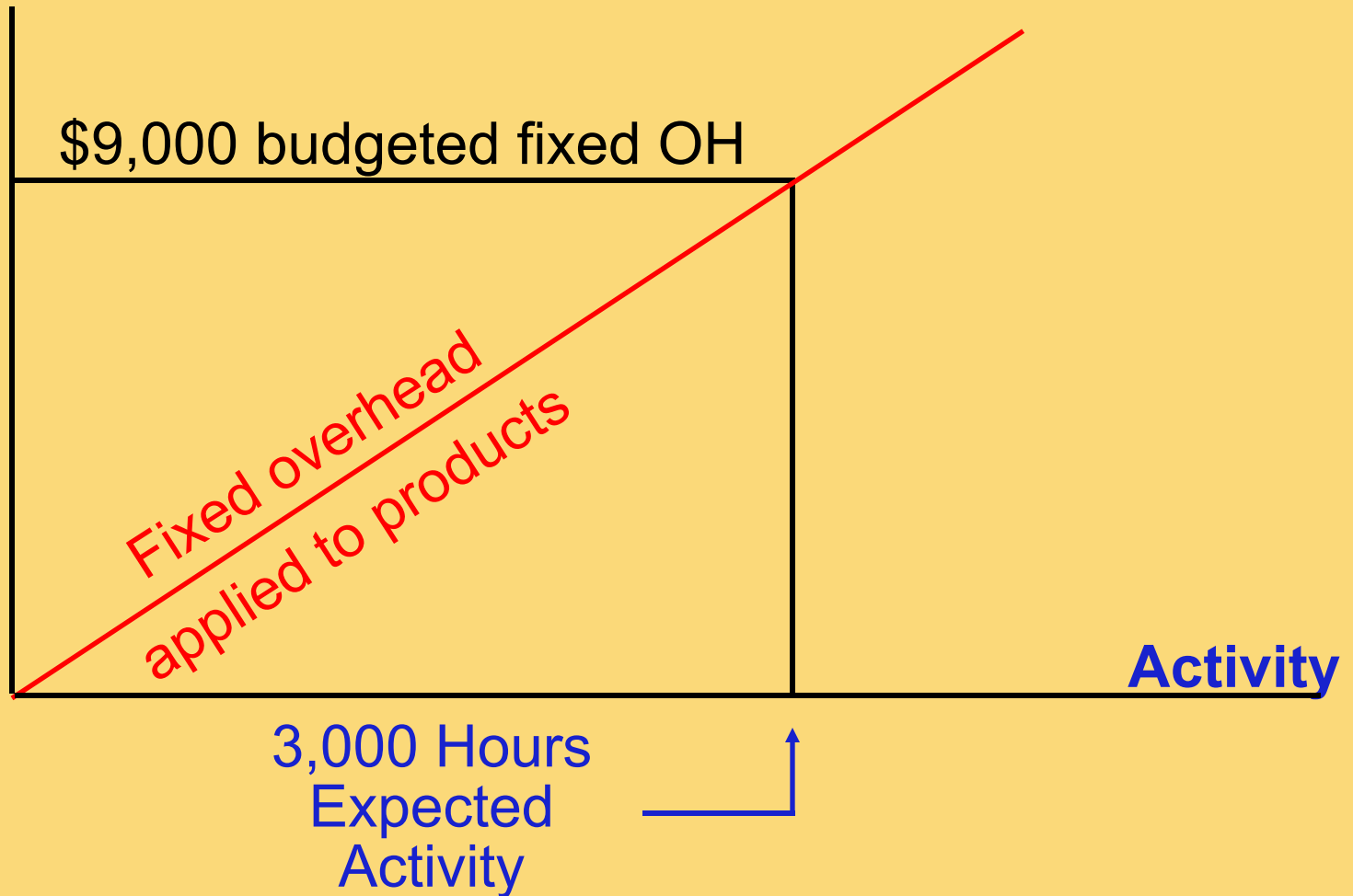
Fixed Overhead Variances – A Graphic Approach



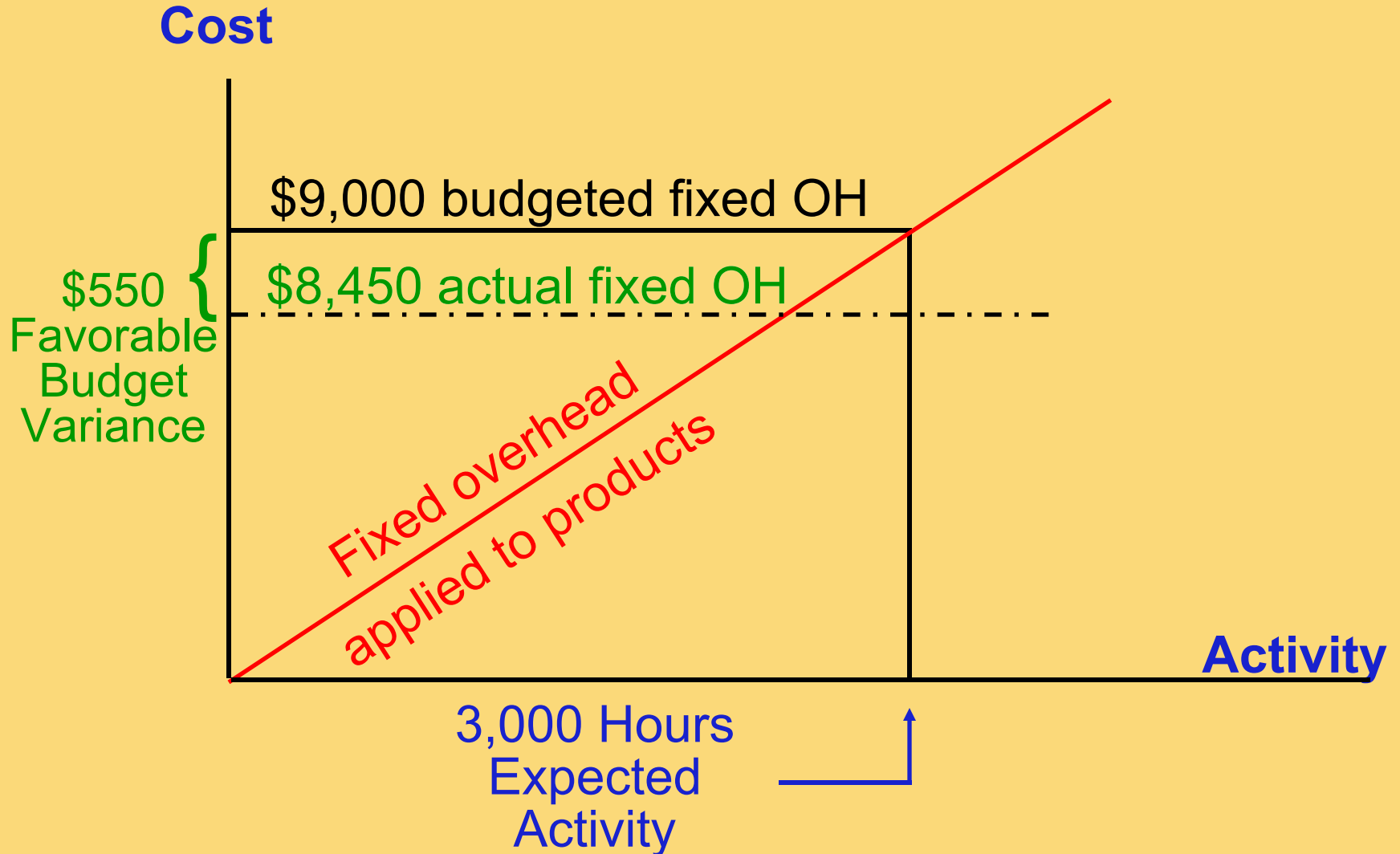
Fixed Overhead Variances – A Graphic Approach



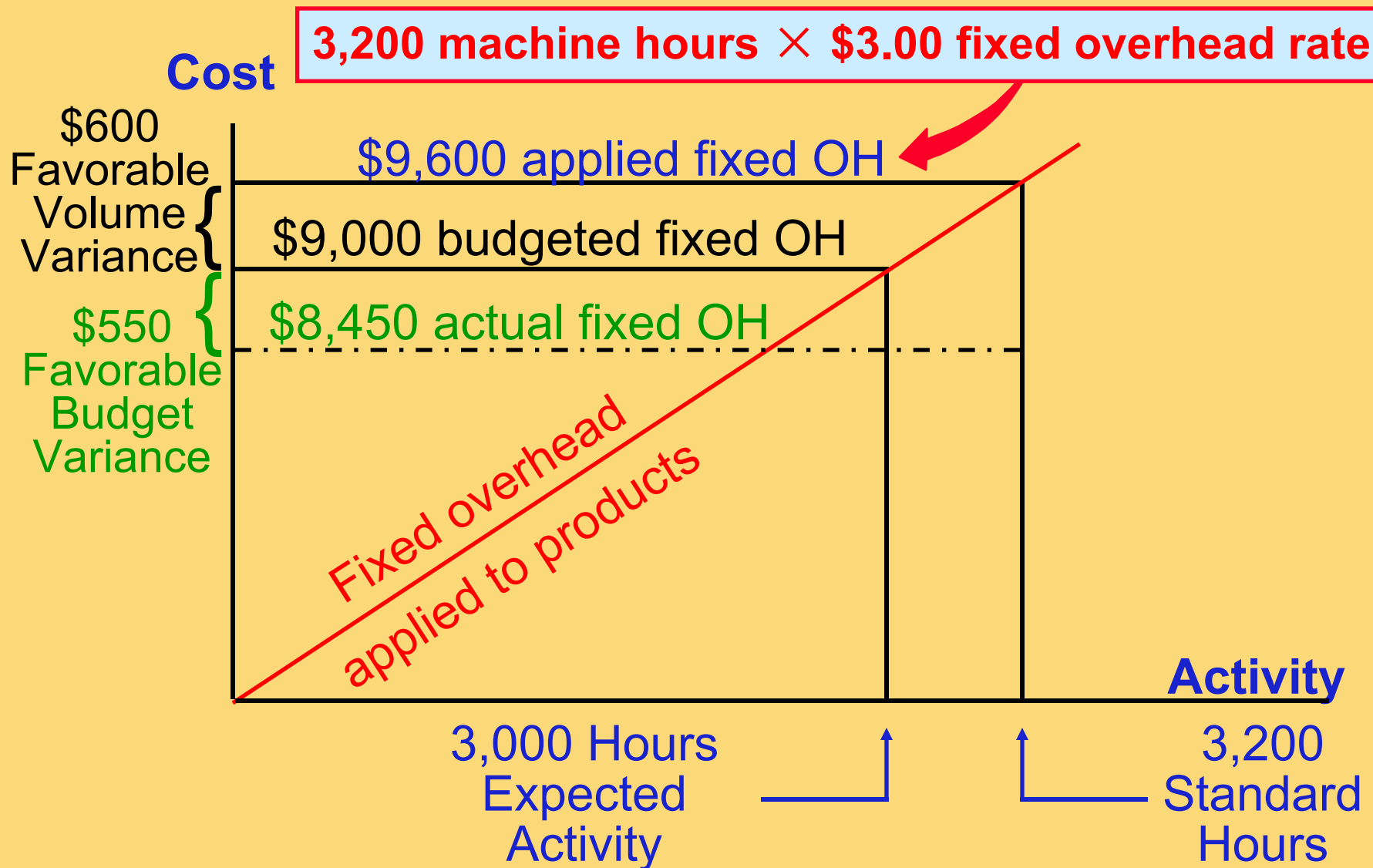
Cost



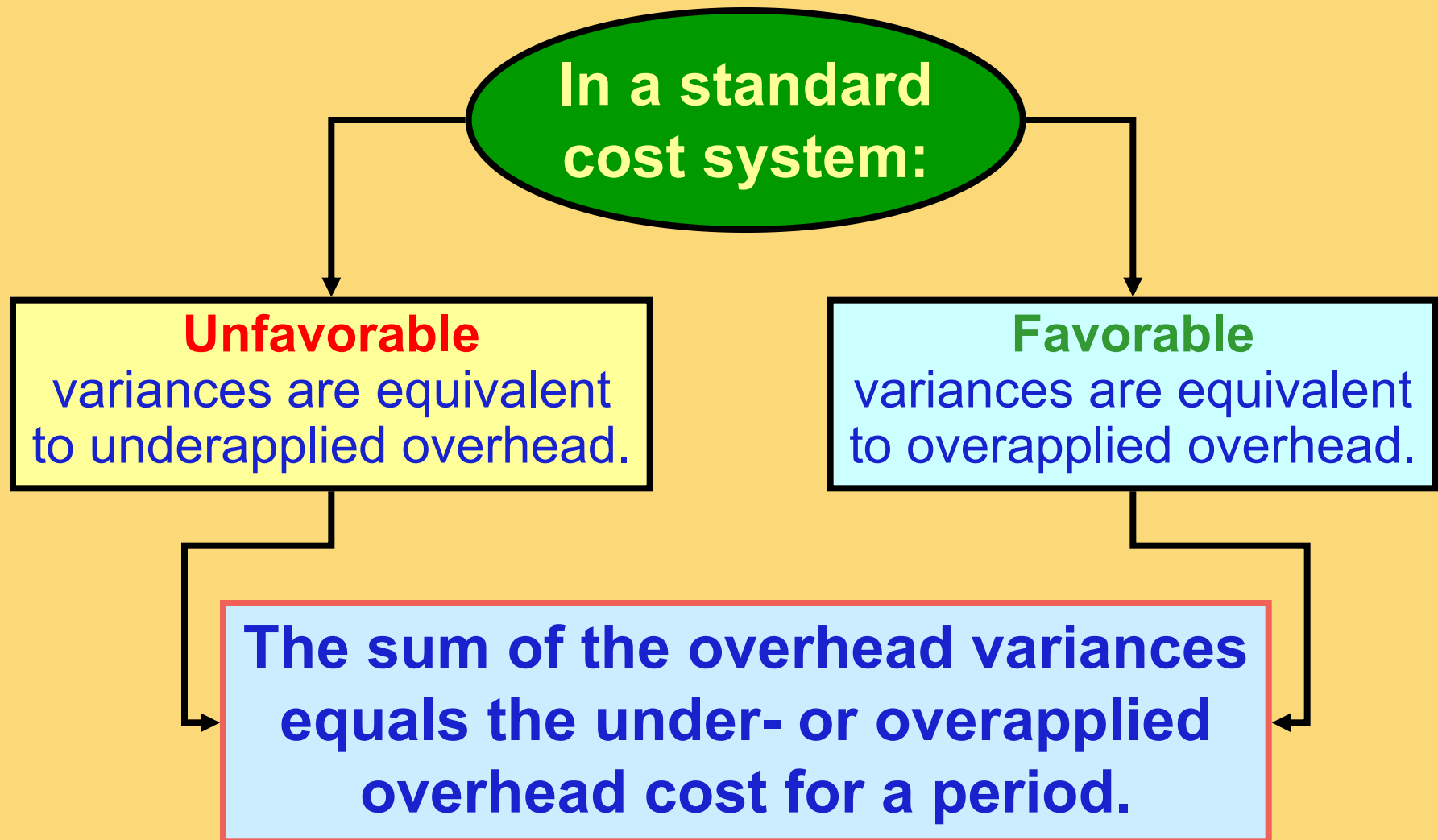
Fixed Overhead Variances – A Graphic Approach



Fixed Overhead Variances – A Graphic Approach



Overhead Variances and Under- or Overapplied Overhead Cost



End of Chapter 11

